Critical Manuscript Review

Design and Benchmarking of an HPC-Accelerated Automated Workflow for Virtual Screening of Thai Traditional Herbal Asthma Formulations

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Reviewer: Neal M. Davies (co-author review)

Executive Summary

This manuscript presents an HPC-accelerated automated workflow for multi-target virtual screening of phytochemicals in a Thai traditional anti-asthma formulation, benchmarked on the LANTA supercomputer. The **core engineering contribution**—workflow automation, SLURM job-array orchestration, and performance benchmarking—is solid and timely. The paper is a reasonable fit for Wiley *Applied Research*, which spans Engineering and Applied Biology.

However, the draft has **significant issues** that must be resolved before submission:

- **7 critical errors** including placeholder text, broken sentences, and an invalid scientific comparison
- **6 major scientific gaps** including missing re-docking validation and unfair baseline comparison
- **20+ English grammar/style errors** ranging from typos to garbled sentences
- **Structural issues** including wrong cross-references and inconsistent formatting

Verdict: The manuscript requires **substantial revision** before submission. With the fixes outlined below, it has a credible chance at *Applied Research*. Submitting in the current state risks immediate desk rejection.

Part A: Journal Fit Assessment

Wiley *Applied Research* (Primary Target)

Scope	Multidisciplinary: Materials, Applied Physics, Chemistry, Applied Biology, Food Science, **Engineering**. Aims to unite fundamental studies with applications.
Fit	**GOOD** — The paper's core contribution is *workflow engineering* applied to a biological problem. HPC benchmarking, SLURM orchestration, and scalability analysis sit at the Engineering ∩ Applied Biology intersection.
Risk	The paper currently leans too heavily into drug-discovery discussion (Section 3.5), which may confuse editors about the primary contribution. Needs sharper engineering framing.

EASR (Backup Target)

Scope	Computer Science Applications (Q3), Engineering miscellaneous (Q3), SJR 0.294
Fit	**GOOD** — More forgiving venue; Thai-based journal familiar with LANTA context.

Strategic Recommendation

For *Applied Research*: frame the paper as "*an engineering solution validated on a biological use case*" rather than "*a drug-discovery study accelerated by HPC*." The drug-discovery results in Section 3.5 should be clearly subordinated as an "Application Case Study."

Part B: English Writing Issues

B1. Critical Errors (Must Fix)

CRITICAL

These errors will cause immediate negative impressions with reviewers and editors.

- **Placeholder text (Section 2.2)** — "The database and libraries employed in this research included **X, Y and Z**." Placeholder left in the manuscript. *Must be filled with actual content.*
- **Typo: "workfkow" (Section 3.1, Para 181)** — "the HPC-automated **workfkow**" → "workflow"
- **Typo: "literate review" (Section 3.1, Para 173)** — "validation through **literate** review" → "**literature** review"
- **Typo: "purposed" (Section 3.4, Para 217)** — "our **purposed** HPC workflow" → "our **proposed** HPC workflow"
- **Double word (Section 3.3, Para 201)** — "resulting **in in** substantially improved" → remove duplicate "in"

- **Broken sentence in Abstract** — "the HPC workflow was expandable for high-throughput screening **reached** 10^6 ligand-protein docking pairs" → completely garbled. Rewrite: "the HPC workflow was scalable to high-throughput screening of up to 10^6 ligand–protein docking pairs"
- **Unintelligible sentence (Section 3.4, Para 217)** — "Such effective scalabilities emerge from the embarrassing parallel structure retrieving ligand-receptor for docking" → must be rewritten entirely. Suggested: "This scalability arises from the embarrassingly parallel nature of molecular docking, where each ligand–receptor pair can be evaluated independently"

B2. Grammar & Syntax Errors

MAJOR

Errors that impede comprehension or appear unprofessional.

- **Abstract: article error** — "a major challenge for **the** computational drug discovery" → drop "the"
- **Intro Para 3: subject–verb disagreement** — "As the size of dataset increases, both file system pressure and computer load increase proportionally, often **leads** to" → "often **leading** to"
- **Section 2.4.3: broken grammar** — "Compounds were classified using the following Boolean logic statement: 'relevant (True)' or 'non-relevant (False)' **is** at least one- or no gene overlap detected" → completely rewrite: "Compounds were classified as *relevant* (True) if at least one predicted target overlapped with the asthma gene set, or *non-relevant* (False) otherwise."
- **Section 2.6 title** — "Molecular docking workflow **implement**" → "**implementation**"
- **Section 3.3: missing verb** — "which **consistent** with" → "which **is** consistent with"
- **Conclusion: broken infinitive** — "enabled **prioritize** multi-target interaction profiles" → "enabled **the prioritization of** multi-target interaction profiles"
- **Intro Para 1** — "Even a simple polyherbal **compound** results in hundreds" → "polyherbal **formulation** generates hundreds" (a compound is a single molecule, not a mixture)
- **Section 2.6.2** — "**Some of** the protein target crystal structure data **was** obtained" → "Crystal structures for the protein targets were retrieved"
- **Figure 2 caption** — "multiple **computes** nodes" → "**compute** nodes"
- **Intro Para 3: punctuation** — "degrade practical throughput despite nominal hardware capability**.** [7]**.**" → period before bracket is wrong; should be "capability [7]."

B3. Style & Readability

MINOR

Issues that, while not errors, weaken the writing and should be improved.

- **Excessive passive voice** — Nearly every sentence uses passive constructions. Aim for ∼30% active voice: "We designed the workflow to…" rather than "The workflow was designed to…"

- **"In order to" overuse** — Appears ∼15 times. Replace with "to" in most cases. "In order to speed up" → "To accelerate"
- **Redundant phrases** — "A total of 126 tasks" → "126 tasks"; "It is evident that" → delete; "It should be noted that" → delete or replace with direct statement.
- **Inconsistent terminology** — "computer nodes" vs. "compute nodes" vs. "computes nodes"; "computer load" vs. "computational load." Standardize throughout.
- **Figure/Table callout format** — Mixes "(Figure X)", "Fig. X", and "Figure X". Standardize per journal style (likely "Figure X" for Wiley).
- **Wordy constructions** — "The preprocessing of the data was carried out by custom Python scripts with the help of the Biopython library" → "Protein structures were preprocessed using custom Python scripts and the Biopython library" (saves 8 words)
- **Abstract length** — Currently ∼280 words. Wiley *Applied Research* likely expects 150–250 words. Trim scalability projection details and focus on the engineering contribution.

Part C: Scientific Validation Issues

C1. Critical Scientific Problems

CRITICAL

Issues that will likely cause rejection if not addressed.

- **Unfair baseline comparison** — The 118.59-fold speedup compares a **laptop** (AMD Ryzen 5, 6 cores, 4.55 GHz boost) against **126 SLURM cores on LANTA** (AMD EPYC 7543, 2.8 GHz). This conflates hardware upgrade with parallelization benefit. A scientifically fair comparison would be sequential execution on LANTA vs. parallel execution on LANTA. The current comparison inflates the reported speedup by including the clock-speed and memory advantages of the supercomputer.

  Recommendation: Either (a) run the sequential baseline on a single LANTA core and re-report, or (b) clearly acknowledge this as a "practical speedup" comparing a typical researcher's workstation against HPC, not a measure of parallelization efficiency per se.

- **Invalid CPU-vs-GPU claim (Section 3.3, Para 207)** — The paper states: "the CPU-based SLURM job array method reported in this work showed a 118-fold acceleration, suggesting that CPU parallelization is just as efficient as GPU acceleration." This comparison is **scientifically invalid** because:

  - Tang et al.'s 50x GPU speedup compares GPU vs. CPU *on the same machine*
  - This paper's 118x compares a laptop vs. a supercomputer
  - The baselines are fundamentally different

  Recommendation: Remove this claim entirely, or reframe as: "While direct comparison is not possible due to differing baselines, the results demonstrate that CPU-based task parallelism via SLURM offers a practical alternative to GPU acceleration for embarrassingly parallel docking workloads."

- **No re-docking validation** — Standard practice in molecular docking studies is to validate the protocol by re-docking a co-crystallized ligand and measuring RMSD (<2 Å is the accepted threshold). This validation is conspicuously absent. Without it, the reliability of the docking scores cannot be assessed.

  Recommendation: Perform re-docking for at least the key targets (PTGS2, ALDH1A1, ADRB2) and report RMSD values, or add as an explicit limitation.

- **Wrong citation: Reference [13]** — Referenced as justification for "automated Python-based informatics workflow," but Ref [13] is Shajii et al. (2021), "A Python-Based *Programming Language* for High-Performance Computational Genomics" (the Seq language). This is not the authors' workflow. Either cite the correct source or remove the citation.

C2. Major Scientific Concerns

MAJOR

Issues that reviewers will likely raise and that weaken the paper's credibility.

- **AutoDock Vina version 1.1.2 is outdated** — Version 1.2.x has been available since 2021 with significant scoring function improvements and multi-threading support. Reviewers will ask why the older version was used. Add a justification (e.g., compatibility with existing scripts, validated performance) or consider updating.

- **Default exhaustiveness parameter** — Exhaustiveness = 8 is the AutoDock Vina default. For a benchmarking paper claiming computational rigor, using the default without justification appears lazy. At minimum, acknowledge this choice and note that higher exhaustiveness would increase both accuracy and runtime proportionally.

- **Grid box dimensions** — "The dimensions of each of the grid maps were kept constant" — if truly constant across all 14 proteins of different sizes, this is methodologically questionable. Active sites vary in volume; a one-size-fits-all grid may miss binding pockets or include excessive solvent space.

- **Homology model quality unreported** — NR1C2 and ELANE were modeled by homology, but the paper reports no template identity, no QMEA/DOPE scores, no Ramachandran plot validation, and no model quality metrics. This is a reproducibility gap.

- **"Novel computational prediction" language** — The claim that solasodine–ALDH1A1 interaction is a "novel computational prediction" because "no previous studies have reported this interaction" is an absence-of-evidence argument. Rephrase to: "To our knowledge, this interaction has not been previously reported in the computational or experimental literature, and warrants further investigation."

- **Scalability projection limitations** — The theoretical projection to 10^6 pairs assumes constant parallel efficiency, but SLURM job arrays have practical limits (typically `MaxArraySize` = 1001–10001). At 10^6 tasks, filesystem metadata pressure, scheduler queuing delays, and Lustre contention would significantly degrade actual performance. These practical HPC engineering constraints should be discussed.

C3. Missing Scientific Elements

- **No ADMET/drug-likeness filtering** — Standard CADD workflows include Lipinski's Rule of Five or similar filters. Adding this for the 9 phytochemicals would strengthen the drug-discovery component with minimal effort.

- **No binding interaction analysis** — The docking poses (Figure 8) are described but not analyzed. Which residues form hydrogen bonds? What hydrophobic contacts stabilize the complex? This is expected in any docking study.
- **No MD simulation or MM-GBSA rescoring** — Acknowledged as a limitation, but strengthens the argument that the paper's contribution is engineering, not drug discovery.
- **Protein preparation is minimal** — No mention of handling missing residues/loops, histidine protonation states, or disulfide bonds. "Addition of polar hydrogens" and "Gasteiger charges" alone is below current best practices.

Part D: Structural & Formatting Issues

D1. Cross-Reference Errors

- **Section 2.11** — References "identical ligand preprocessing scheme as outlined in **Section 2.5**" but ligand preparation is in **Section 2.6.1**. Section 2.5 covers pathway-based filtering.
- **Phase labeling inconsistency** — Phase I = Section 2.4, Phase II = Section 2.5, Phase III = Section 2.6. But Section 2.3 (hardware) sits between the overview and Phase I, breaking the logical flow. Consider reordering.

D2. Reference Format

- **Non-standard citation format** — The references use a narrative format (author, title, journal, volume, pages, DOI). Wiley *Applied Research* uses numbered references in Vancouver-like style. The entire reference list will need reformatting.
- **Future-dated references** — References [27] (2026) and [42] (2026) carry dates that may not yet be published at the time of submission. Verify publication status and use "in press" or "early view" as appropriate.

D3. Figure & Table Captions

- **Inconsistent caption punctuation** — "FIGURE 5." has a period after the number; others do not. Standardize.
- **Caption length** — Some captions are very long (>3 lines). Consider shortening to one-sentence descriptions with details moved to the main text.

Part E: Revision Priority Matrix

#	Severity	Issue	Effort
1	CRITICAL	Fill placeholder "X, Y and Z" in Section 2.2	Low
2	CRITICAL	Fix 5 typos (workfkow, literate, purposed, computes, in in)	Low
3	CRITICAL	Rewrite 2 garbled sentences (Abstract, Para 217)	Low
4	CRITICAL	Remove/qualify CPU-vs-GPU comparison claim	Low
5	CRITICAL	Acknowledge unfair laptop-vs-HPC baseline	Low

#	Severity	Issue	Effort
6	CRITICAL	Fix wrong citation (Ref [13])	Low
7	CRITICAL	Fix cross-reference: Section 2.5 → 2.6.1	Low
8	MAJOR	Add re-docking validation (or limitation note)	Med
9	MAJOR	Justify AutoDock Vina 1.1.2 choice	Low
10	MAJOR	Fix 10 grammar/syntax errors (Part B2)	Med
11	MAJOR	Report homology model quality metrics	Med
12	MAJOR	Temper "novel prediction" language	Low
13	MAJOR	Discuss SLURM scalability limits (MaxArraySize)	Low
14	MAJOR	Reformat references to Wiley style	Med
15	MINOR	Reduce passive voice (∼30% active target)	Med
16	MINOR	Replace "in order to" globally	Low
17	MINOR	Trim abstract to <250 words	Low
18	MINOR	Standardize figure/table callouts	Low
19	MINOR	Add drug-likeness assessment (Lipinski)	Med
20	MINOR	Add binding interaction analysis for key hits	Med

Part F: Positive Aspects

The manuscript has genuine strengths that should be preserved and highlighted:

1. **Clear workflow architecture** — The three-phase design (data acquisition → target prioritization → HPC docking) is logically sound and well-structured.
2. **Reproducible methodology** — The use of public databases (PubChem, ChEMBL, KEGG, PDB) and open-source tools (RDKit, Open Babel, AutoDock Vina) supports reproducibility.
3. **Practical HPC benchmarking** — The performance metrics (speedup, parallel efficiency, throughput, numerical consistency) are comprehensive and well-chosen.
4. **Real infrastructure** — Using LANTA with actual SLURM job arrays gives the paper practical credibility over purely simulated benchmarks.
5. **Honest scalability discussion** — The theoretical projection is clearly labeled as such, with assumptions stated.
6. **Relevant case study** — Thai traditional medicine + HPC is a timely and underexplored intersection.
7. **Strong data reduction narrative** — The 92.9% filtering from 197 to 14 targets is a compelling preprocessing result.

Verdict: With the critical and major issues addressed, this manuscript presents a solid contribution at the intersection of HPC engineering and computational pharmacology. The workflow-engineering angle is the paper's strongest asset and should be foregrounded for *Applied Research*.

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